INCH-POUND

MIL-C-24643/58D
22 November 1994
SUPERSEDING
MIL-C-24643/58C
14 March 1994

MILITARY SPECIFICATION SHEET

CABLE, ELECTRICAL, TYPE LS2CS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-24643.

REQUIREMENTS:

Qualification required.

Construction (non-watertight)

First - Copper conductor, tin coated, ASTM B 286, size 26-7. Maximum diameter 0.020 inch.

Second - Separator (opaque white) may be used at manufacturer's option where required to provide free-stripping insulation.

Third - Crosslinked polyethylene insulation. Nominal wall thickness 0.015 inch; wall thickness may vary from nominal as necessary, in order for completed cable to meet the specified electrical requirements. Standard identification code applied by method 6.

Fourth - Two conductors (one black and one white) cabled together to form a pair in accordance with table I.

Fifth - The specified number of pairs (see table II) cabled together in accordance with table III. Cabling sequence shall be consecutive, starting with no. 1, from the center outward. Fillers shall be used as necessary to make a firm, well rounded assembly.

Sixth - Binder tape applied helically with overlap.

Seventh- Braided shield of tin coated copper. Minimum coverage of 85 percent.

Eighth - Spirally wrapped 0.002 inch thick polyethylene terephthalate (mylar) separator tape with 25 percent minimum overlap.

Ninth - Braided shield of tin coated copper. Minimum coverage of 85 percent.

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Tenth - Binder tape applied helically with overlap.
Eleventh - Cross-linked polyolefin jacket. (see table III for thickness)

TABLE I. Pair lay

Left hand lay										
/ / / / /					65	66	67	62,68 71,74,77	63,69 72,75	61,64 70,73, 76
/ / / / /	/ / / /				53	54	55	45,48 51,56,59	43,46, 49 52,56, 60	44,47 50,58
/ / / / /	/ / / /	/ / / /			27,39	28,40	29,41	19,22,25 30,33,36, 42	20,23, 26 31,34, 37	21,24 32,35, 38
/ / / / /	/ / / /	/ / / /	/ / / /		13 1	14 2	15 3	7,10,16 4	8,11,1 7 5	9,12,1 8 6
LS2CS-77	LS2CS-60	LS2CS-42	LS2CS-18	LS2CS-6	1.6	3.5	2.5	4.5	2.1	2.9
					LENGTH OF LAY (INCHES)					

PAIR NUMBER

TABLE II. Details.

	Military part no. M24643/58	Type and size	Number of pairs	Cable jacket thickness min avg (inch)	Dia over shielded core (inch nom.)	Overall diameter (inch	
						min	max
*	-01UD	LS2CS-6	6	0.063	0.300	0.40	0.43
*	-02UD	LS2CS-18	18	.063	.435	.55	.59
*	-03UD	LS2CS-42	42	.063	.610	.75	.80
*	-04UD	LS2CS-60	60	.082	.790	.88	.93
*	-05UD	LS2CS-77	77	.085	.860	1.00	1.07

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TABLE III. CABLE LAY

	LS2CS-6		LS2CS-18		LS2CS-42		LS2CS-60		LS2CS-77	
	No. of pairs	Lay (inches)	No. of pairs	Lay (inches)	No. of pairs	Lay (inches)	No. of pairs	Lay (inches)	No. of pairs	Lay (inches)
Core	6	3.50	6	3.50	1	1.60	6	3.50	3	2.75
Layer 1	-	--	12	6.00	7	4.00	12	6.50	9	5.00
Layer 2	-	--	-	--	14	6.50	18	8.00	15	7.50
Layer 3	-	--	-	--	20	8.50	24	10.00	22	10.00
Layer 4	-	--	-	--	-	--	-	--	28	12.00

EXAMINATION AND TESTS:

	Requirements
Basic electrical:	
Conductor Resistance - ohms/1000 feet at 25°C, maximum	42.17
Voltage withstand - volts, root mean square, minimum	
Conductor to conductor	2000
Insulation resistance - Megohms/1000 feet minimum	
Conductor to shield	500
Shield to Shield (outer shields only)	100
Jacket flaws	No failure
Conductor and shield continuity	No failure
Group A:	
Visual and dimensional	No failure
Characteristic impedance - at 1MHz, ohms	135 ± 10
Group B:	
Cross-linked proof test (percent, maximum)	
Insulation	50
Jacket (when tested at 200°C)	50
Cold bending, cable - mandrel diameter, 12 times cable diameter	No damage
Bending endurance - at 25 ± 2°C, cycles, minimum	25
Tear - pounds per inch thickness, minimum (ASTM D 470)	35
Physicals (unaged)	
Insulation	
Tensile strength - lb/in^2, minimum	1800
Elongation - percent, minimum	250
Jacket (cable)	
Tensile strength - lb/in^2, minimum	1300
Elongation - percent, minimum	160

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EXAMINATION AND TESTS:

	Requirements
Group C:	
Physicals (aged)	
Insulation	
Air Oven	
Tensile strength - percent of unaged, minimum	80
Elongation - percent of unaged minimum	80
Jacket (cable)	
Air oven	
Tensile strength - percent of unaged, minimum	60
Elongation - percent of unaged minimum	60
Hot oil immersion	
Tensile strength - percent of unaged, minimum	50
Elongation - percent of unaged minimum	50
Shrinkage	No failure
Heat distortion - percent of unaged, maximum	30
Permanence of printing (jacket) - cycles, minimum	125
* Shield - conformance to material, construction and coverage	No failure
Surface transfer impedance	
Milliohms per meter, maximum	700
Emp response time - db, minimum	60
Group D:	
Flame propagation (cable)	No failure

QUALIFICATION INSPECTION:

Qualification inspection shall include basic electrical, all of groups A, B, C and D, plus the following:

	Requirements
Aging and compatibility (cable) (125 ± 5°C)	No failure
Abrasion resistance (jacket) - scrapes, minimum	75
Acid gas equivalent - percent, maximum	
Jacket	2
Fillers	2
Insulation	18
Shield insulation	2
Halogen content - percent, maximum	
Jacket	0.2
Fillers	0.2
Immersion (jacket)	
Tensile strength - percent of unaged, minimum	50

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QUALIFICATION INSPECTION:

	Requirements
Elongation - percent of unaged, minimum	50
Smoke index, maximum	
Jacket	25
Fillers	45
Insulation	45
Toxicity index, maximum	
Jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) - (type A) hardness, minimum	80
Weathering (jacket)	No failure

UNIT ORDERING LENGTHS: All sizes 1000 feet (nominal).

NOTE: The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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